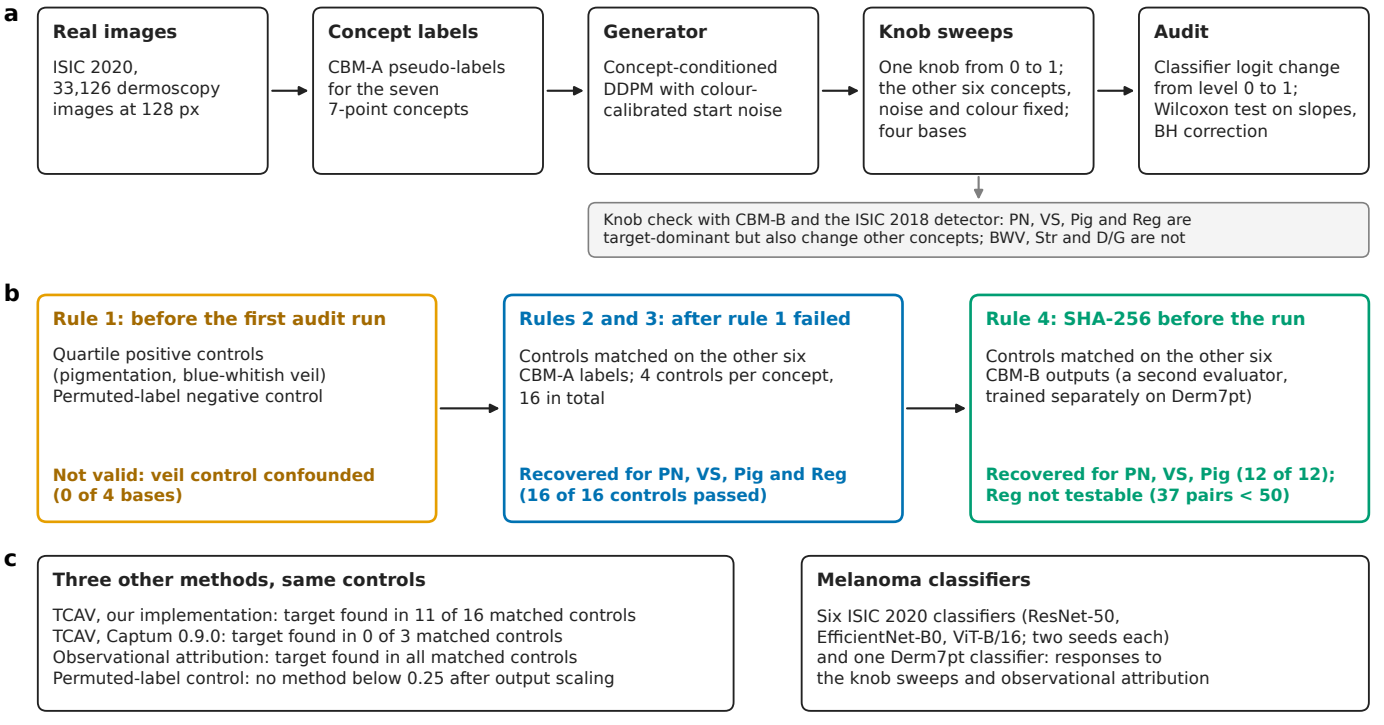
a
Real images
ISIC 2020,
33,126 dermoscopy
images at 128 px
Concept labels
CBM-A pseudo-labels
for the seven
7-point concepts
Generator
Concept-conditioned
DDPM with colour-
calibrated start noise
Knob sweeps
One knob from 0 to 1;
the other six concepts,
noise and colour fixed;
four bases
Audit
Classifier logit change
from level 0 to 1;
Wilcoxon test on slopes,
BH correction
Knob check with CBM-B and the ISIC 2018 detector: PN, VS, Pig and Reg are
target-dominant but also change other concepts; BWV, Str and D/G are not
b
Rule 1: before the first audit run
Quartile positive controls
(pigmentation, blue-whitish veil)
Permuted-label negative control
Not valid: veil control confounded
(0 of 4 bases)
Rules 2 and 3: after rule 1 failed
Controls matched on the other six
CBM-A labels; 4 controls per concept,
16 in total
Recovered for PN, VS, Pig and Reg
(16 of 16 controls passed)
Rule 4: SHA-256 before the run
Controls matched on the other six
CBM-B outputs (a second evaluator,
trained separately on Derm7pt)
Recovered for PN, VS, Pig (12 of 12);
Reg not testable (37 pairs < 50)
c
Three other methods, same controls
TCAV, our implementation: target found in 11 of 16 matched controls
TCAV, Captum 0.9.0: target found in 0 of 3 matched controls
Observational attribution: target found in all matched controls
Permuted-label control: no method below 0.25 after output scaling
Melanoma classifiers
Six ISIC 2020 classifiers (ResNet-50,
EfficientNet-B0, ViT-B/16; two seeds each)
and one Derm7pt classifier: responses to
the knob sweeps and observational attribution